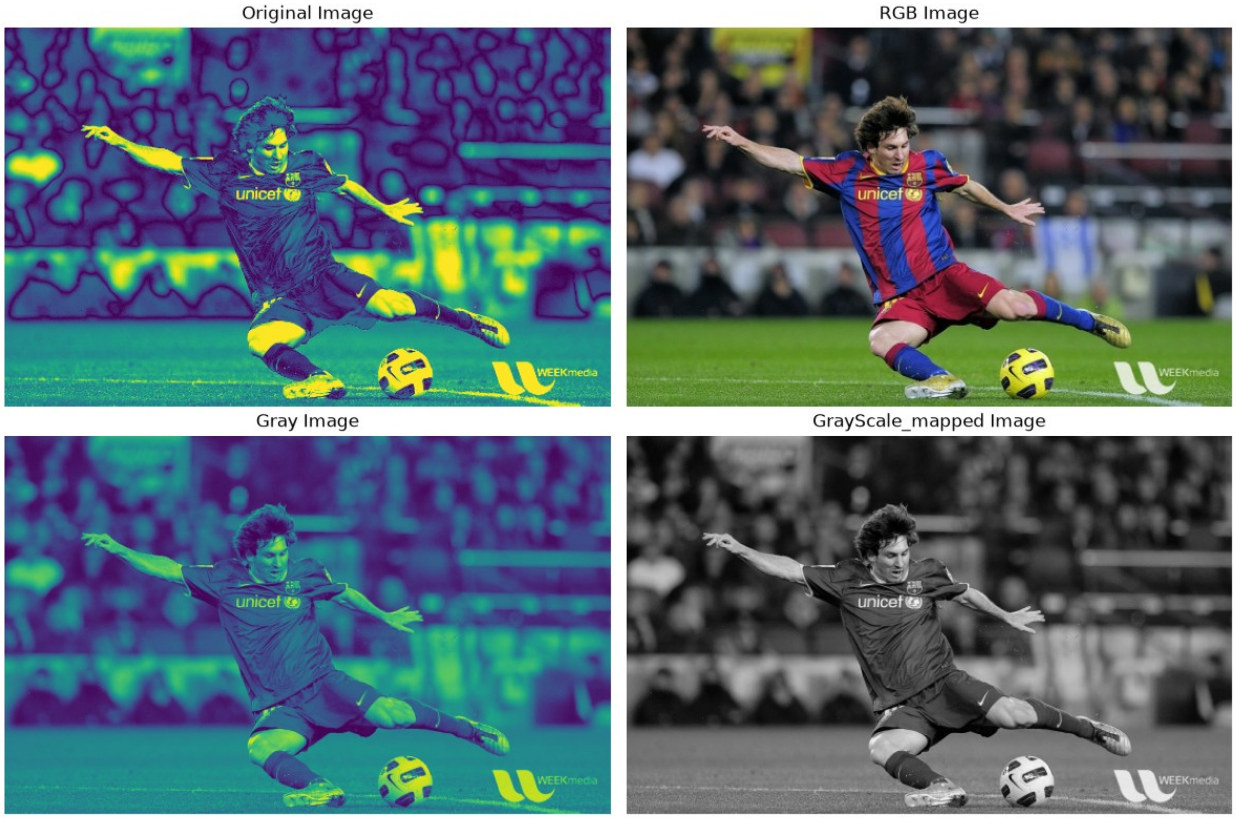
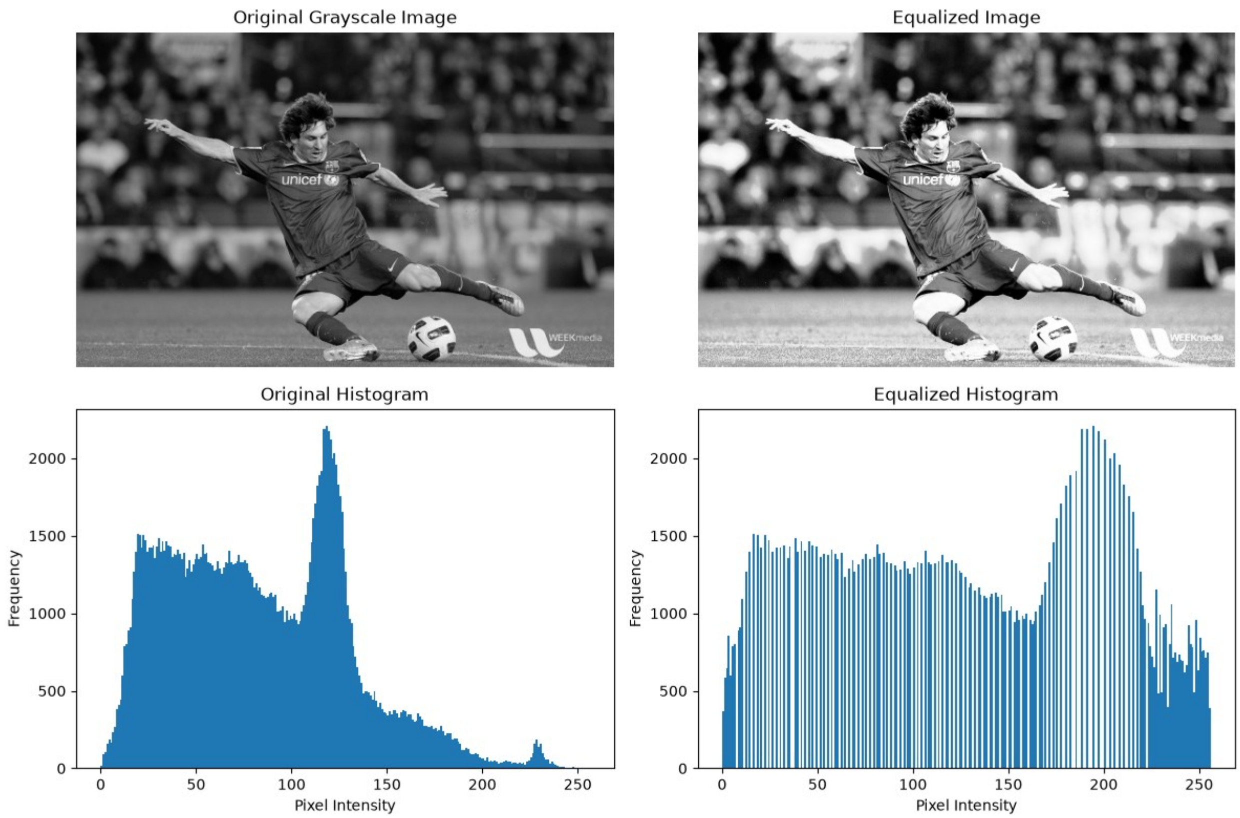
```python
import cv2
import matplotlib.pyplot as plt

img = cv2.imread('images/data/messi5.jpg')

gray_image = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

rgb_image = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(12, 8))


plt.subplot(2, 2, 1)
plt.imshow(img)
plt.title("Original Image")
plt.axis("off")

plt.subplot(2, 2, 2)
plt.imshow(rgb_image)
plt.title("RGB Image")
plt.axis("off")

plt.subplot(2, 2, 3)
plt.imshow(gray_image)
plt.title("Gray Image")
plt.axis("off")

plt.subplot(2, 2, 4)
plt.imshow(gray_image, cmap="gray")
plt.title("GrayScale_mapped Image")
plt.axis("off")

plt.tight_layout()
plt.show()
```



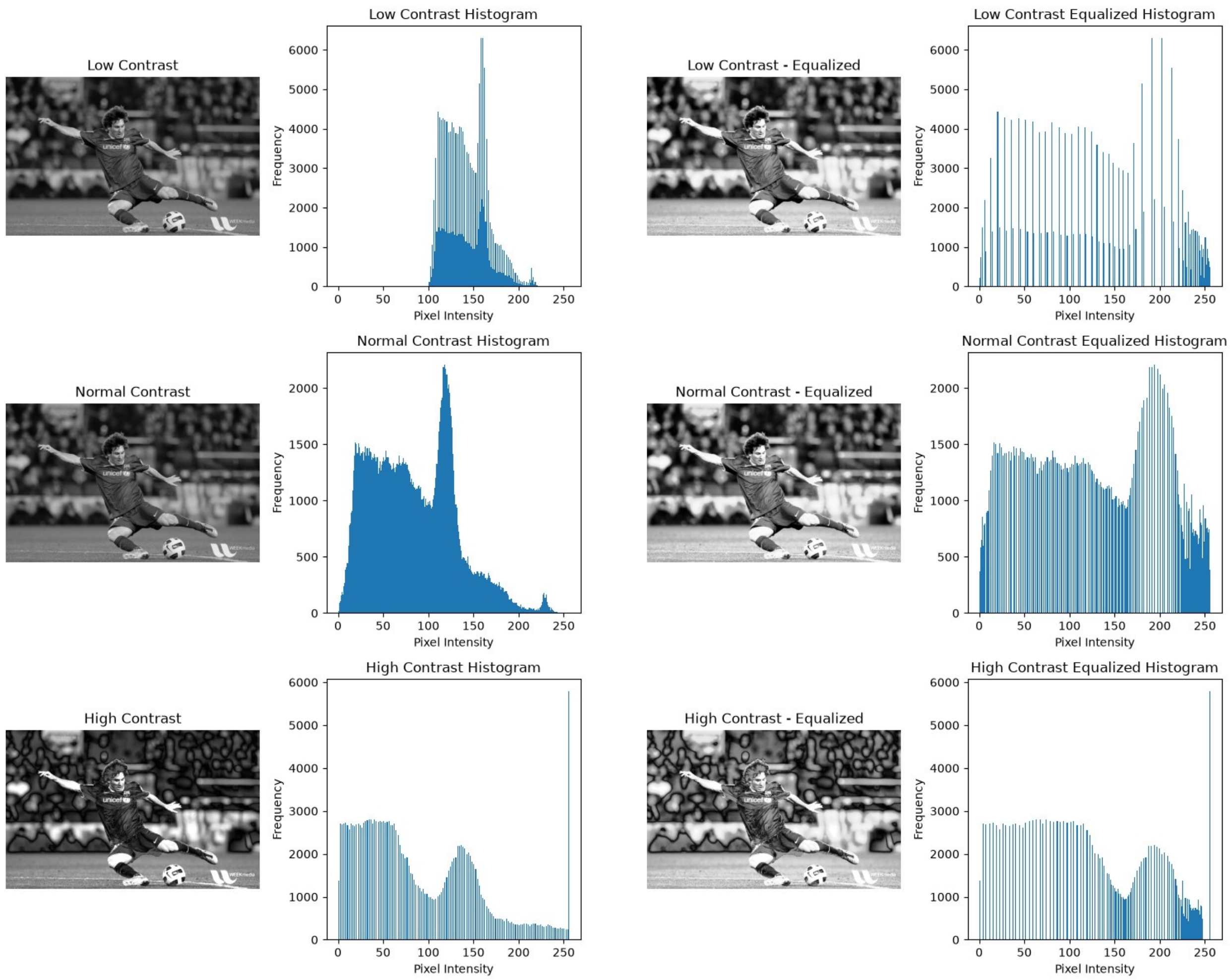

```
equalized_image = cv2.equalizeHist(gray_image)

plt.figure(figsize=(12, 8))

plt.subplot(2, 2, 1)
plt.imshow(gray_image, cmap="gray")
plt.title("Original Grayscale Image")
plt.axis("off")

plt.subplot(2, 2, 2)
plt.imshow(equalized_image, cmap="gray")
plt.title("Equalized Image")
plt.axis("off")

# 5. Histogram of original image
plt.subplot(2, 2, 3)
plt.hist(gray_image.ravel(), bins=256, range=[0, 256])
plt.title("Original Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

# 6. Histogram of equalized image
plt.subplot(2, 2, 4)
plt.hist(equalized_image.ravel(), bins=256, range=[0, 256])
plt.title("Equalized Histogram")
```

```python
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```




```python
low_contrast = cv2.convertScaleAbs(gray_image, alpha=0.5, beta=100)
normal_contrast = gray_image
high_contrast = cv2.convertScaleAbs(gray_image, alpha=2.0, beta=-100)

images = [
    ("Low Contrast", low_contrast),
    ("Normal Contrast", normal_contrast),
    ("High Contrast", high_contrast)
]

plt.figure(figsize=(15, 12))

for i, (name, img) in enumerate(images):

    # Original contrast image
    plt.subplot(3, 4, i * 4 + 1)
    plt.imshow(img, cmap="gray")
    plt.title(name)
    plt.axis("off")
```

```python
    # Original histogram
    plt.subplot(3, 4, i * 4 + 2)
    plt.hist(img.ravel(), bins=256, range=[0, 256])
    plt.title(name + " Histogram")
    plt.xlabel("Pixel Intensity")
    plt.ylabel("Frequency")

    # Histogram equalization
    equalized = cv2.equalizeHist(img)

    # Equalized image
    plt.subplot(3, 4, i * 4 + 3)
    plt.imshow(equalized, cmap="gray")
    plt.title(name + " - Equalized")
    plt.axis("off")

    # Equalized histogram
    plt.subplot(3, 4, i * 4 + 4)
    plt.hist(equalized.ravel(), bins=256, range=[0, 256])
    plt.title(name + " Equalized Histogram")
    plt.xlabel("Pixel Intensity")
    plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```


Low Contrast
Low Contrast Histogram
Low Contrast - Equalized
Low Contrast Equalized Histogram
Normal Contrast
Normal Contrast Histogram
Normal Contrast - Equalized
Normal Contrast Equalized Histogram
High Contrast
High Contrast Histogram
High Contrast - Equalized
High Contrast Equalized Histogram
Frequency
Pixel Intensity